

EE1080 Probability, Quiz 7

27th April, 2026

Max. Marks: 12. Time: 1 hour.

Instructions

- Please **write your roll number and course id** prominently in the first page of the answer sheet.
- No laptops, mobile devices etc. allowed.
- please write supporting arguments for any of the statements you make. any result proved in the lectures can be used by stating it clearly, no need to prove them in your answer.

1. (2+2) Which of the following matrices can be covariance matrices:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

For the cases where the matrices are covariance matrices, find the correlation between the two random variables of the random vector.

2. (2+2) Describe p.d.f of Gaussian vectors with the following covariance matrices:

$$K_X = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}, K_Y = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix},$$

Can you generate samples of Y suppose from samples of X and can you generate samples of X from Y ? Describe how to do it if whenever its possible.

3. (2+2) Let $V = (X, Y_1, Y_2)^T$ be a zero mean Gaussian vector described through the covariance matrix K_V . Assume K_V is invertible and that the entries are as shown below:

$$K_V = \begin{bmatrix} \sigma_x^2 & \rho_1 \sigma_x \sigma_{y_1} & \rho_2 \sigma_x \sigma_{y_2} \\ \rho_1 \sigma_x \sigma_{y_1} & \sigma_{y_1}^2 & \rho \sigma_{y_1} \sigma_{y_2} \\ \rho_2 \sigma_x \sigma_{y_2} & \rho \sigma_{y_1} \sigma_{y_2} & \sigma_{y_2}^2 \end{bmatrix}$$

- (a) Find the LMMSE estimator of X from Y_1, Y_2 .
(b) Find the MSE error of the MMSE estimator.